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ABSTRACT

Ion channels are proteins that play a significant role in physiological processes, including neuronal excitability and signal transduction. However, the precise mechanisms by which these proteins facilitate ion diffusion through cell membranes are not well understood. This is because experimental techniques to characterize ion channel activity operate on a time scale too large to understand the role of the various protein conformations on diffusion. Meanwhile, computational approaches operate on a time scale too short to rationalize the observed behavior at the microscopic scale. In this paper, we present a continuous-time random walk model that aims to bridge the scales between the atomistic models of ion channels and the experimental measurement of their conductance. We show how diffusion slows down in complex systems by using 3D lattices that map out the pore geometry of two channels: Na_v1.7 and gramicidin. We also introduce spatial and dynamic site disorder to account for system heterogeneity beyond the mean field approximation. Computed diffusion coefficients show that an increase in spatial disorder slows down diffusion kinetics, while dynamic disorder has the opposite effect. Our results imply that microscopic or phenomenological models based on the potential of mean force data overlook the functional importance of protein dynamics on ion diffusion through channels.

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I. INTRODUCTION

Lipid bilayers are impermeable to ions, which leaves embedded proteins solely responsible for ion transport through cell membranes.^{1,2} Ion channels are a subgroup of these transport proteins that create a pore spanning the membrane (i.e., a channel), filled with water, through which ions can diffuse (Fig. 1).^{3,4} Like all proteins, ion channels adopt many conformations that modulate the kinetics of ion diffusion through the pore.^{5–7} For example, protein side chain rotations, loop dynamics, and the motions of charged and polarizable groups on the pore walls are expected to have a significant effect on ion diffusion kinetics. For some channels, there exist different states, each consisting of ensembles of the aforementioned conformations. Some of these states, such as the closed and inactivated states, completely suppress ion flow through the channel,^{6,8,9} while others, such as the open state, let the ions diffuse through

the pore. Variations in the physicochemical properties of the environment drive transitions between conformational states, thereby modulating the permeability of cell membranes over time.^{6,10,11} For example, voltage-gated sodium channels (Na_vs) are activated by changes in the transmembrane voltage of excitable cells.^{6,8,11,12} When open, Na_vs enable Na⁺ flow inside the cells, causing the rising phase of action potentials.^{8,13,14}

Ion channels have been extensively studied, both experimentally and theoretically. Experimentally, we can measure membrane conductance with patch-clamp techniques.^{15,16} Patch-clamp recordings can even be performed on membrane sections containing a single ion channel.^{15,17,18} Yet, even these single channel measurements are not resolved on time and length scales large enough to allow the quantification of the many molecular factors that can shape ion diffusion kinetics. Theoretically, the prediction of ion diffusion kinetics inside ion channels remains an unsolved problem.^{4,19–21} In

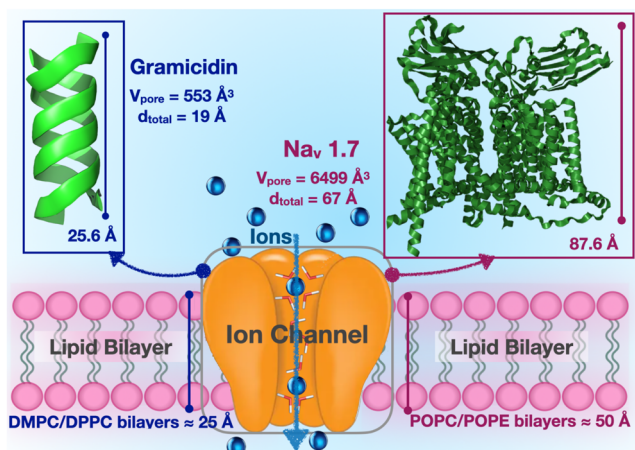


FIG. 1. Ion channels are proteins whose dynamics enable ion diffusion through cell membranes. In this paper, we present models of the gramicidin and $\text{Na}_v 1.7$ pores.

the past, ion channel permeation has been modeled with Brownian dynamics, where the motion of a charged particle is governed by the Langevin equation.^{21–32} Although continuum and kinetic Monte Carlo methods were used to solve Brownian dynamics in early studies of ion channels, the system anisotropy and the lack of molecular resolution severely limit the application and impact of these methods.^{10,21,34–40}

Kinetic models were then abandoned in the late 1990s in favor of MD, which offered the potential to resolve channel dynamics at the molecular scale.¹⁰ However, where kinetics models suffer from a lack of molecular structural data, MD lacks functional information. Indeed, even modern MD algorithms can, at best, model single ion permeation events,^{10,39,41–45} which does not provide full statistical sampling; channel conductivity is the property average of, at least, thousands of permeation events. This disparity in time and length scales between MD simulations and measured properties limits our understanding of many of these channels.^{46–55} For example, MD simulations revealed the conductivity-selectivity paradox, which describes the apparent discrepancy between the high conductivity of Na_v s, suggesting fast permeation, and the high selectivity of such channels, suggesting relatively strong binding of the target ions in the selectivity filter.^{10,56–58} Over the past decades, variations on kinetic models have emerged to address the aforementioned issues.^{4,19,27,59–61} Most methods use MD-based parameters incorporated into a potential of mean force (PMF), which is then used to calculate a position-dependent coefficient of diffusion.^{28,29,31,32} This coefficient is then used in Poisson–Nernst–Planck (PNP) equations, giving a profile for the trajectory of the ion. However, these models still rely on the mean field approximation, where Na^+ kinetics are computed from a potential of mean force that averages out dynamical heterogeneity (i.e., the ion moves along an averaged profile of the channel, which is the same across the channel length).

In this paper, we present a Continuous-Time Random Walk (CTRW)^{5,62,63} algorithm to simulate ion diffusion through an ever-evolving medium with direct application to ion channels. We perform these walks on 3D lattices that map out the geometry of real channel pores, as determined experimentally. Here, we present

results for the well-characterized, cylindrical gramicidin pore^{64,65} and the more complex human $\text{Na}_v 1.7$ ^{8,11} pore, as shown in Fig. 1.

We parametrize our CTRW algorithm to systematically investigate the influence of spatial and temporal disorder on ion permeation kinetics through the gramicidin and $\text{Na}_v 1.7$ pores, using an open cubic lattice for control. We do that by introducing a site disorder that results in a variable time step, slowing down (accelerating) ion dynamics when the disorder is high (low). This site disorder aims to mimic the influence of the molecular motions in protein environments that occur on disparate time and length scales, shaping diffusion kinetics. Our results indicate that spatial and temporal disorder significantly affect the diffusion coefficient in all cases investigated, suggesting that mean field approximations and continuum models are not sufficient to model transport through ion channels.

II. METHODS

A. Discretizing the channel pore volume into a 3D lattice for random walks

We run our CTRW algorithm on 3D lattices that map out the geometry of each channel pore. To build such a lattice, we start with a structure in the Research Collaboratory for Structural Bioinformatics (RCSB) Protein Data Bank (PDB). We selected PDB ID 1C4D⁶⁴ and 6J8I⁶⁶ for gramicidin and $\text{Na}_v 1.7$, respectively. We simplify PDB ID 6J8I to only retain the coordinates of the pore helices, which are helices S5 and S6 in each domain.^{66,67} We then orient the structure in PDB ID 1C4D such that the principal axis of the pore is aligned with the z -axis (PDB ID 6J8I is already aligned along that direction) and discretize space into a cubic lattice of $1 \text{ \AA}^3$ sites. This site volume was chosen to match the ionic radius of Na^+ ⁶⁸ in each direction. Each lattice site corresponding to an atomic position listed in the PDB is marked as occupied. The remaining unoccupied sites trace out the voids in the structure, which include the channel pore as well as any gaps between segments or structural domains. To further isolate the channel pore, we select one unoccupied site along the central z -axis on each (xy) plane, marking it as a pore site. We then move radially outward from these points, marking each unoccupied site as a pore site until one occupied site is reached on each plane. Finally, all unoccupied sites in each layer that have more than two pore sites as nearest neighbors are also marked as pore sites. For reference, we also created an ideal, open lattice of 67 pore sites in each direction ($67 \times 67 \times 67 \text{ \AA}^3$), with additional occupied sites forming the boundary. The length of this lattice cube was chosen to match the size of the longest axis for the $\text{Na}_v 1.7$ pore. The 3D lattices obtained are shown in Fig. 2. The average diameters of the gramicidin channel and the three sections of the $\text{Na}_v 1.7$ are shown in pink, and the number of lattice sites along each site index in the z -axis is shown in Fig. S2.

The total sites of the $\text{Na}_v 1.7$ pore are divided into sections along the z -axis based on the residues that line the pore, namely the selectivity filter (SF), central cavity (CC), and inactivation gate (IG) of $\text{Na}_v 1.7$. Based on the demarcations of these sections, similar sections are created in the open lattice and the gramicidin lattice, with the size ratio along the z -axis kept the same (to the nearest pore site). The length of the pore from the start of the SF and the end of the IG is calculated, and an additional $\sim 10\%$ length of the pore is added at both ends of the pore to determine the entry and exit sites of the channel.

TABLE I. Coordinates of the initial point (marked red in Fig. 2) and z-coordinates of the start and finish of the various regions within each lattice, as explained in the text. SF = selectivity filter, CC = central cavity, IG = inactivation gate.

z-coordinate (Å)	Open lattice	Gramicidin	Na _v 1.7
Initial point	(33, 33, 50)	(3, 3, 18)	(83, 76, 78)
Entry into pore	47	16	75
Start of SF	41	14	69
End of SF	37	12	65
Start of CC	37	12	65
End of CC	21	6	49
Start of IG	21	6	49
End of IG	9	4	37
Exit from pore	3	2	31
Pore length (d_{diff})	44	14	44

The demarcations at the end of the regions are the z-coordinates for when the walker is considered to have “entered” or “exited” the channel. The walker starts the walk at a distance that is ~5% of the pore’s original length away from the topmost demarcated line. The coordinates of the initial point and the z-coordinates of the start and finish of the various regions within each lattice are given in Table I.

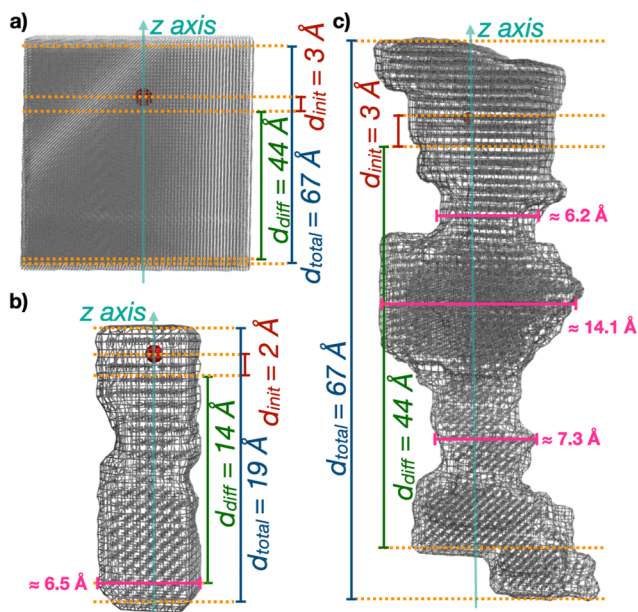


FIG. 2. 3D lattices used for the random walks: (a) open lattice (control), (b) gramicidin, and (c) Na_v1.7. Each site has a volume of $1 \times 1 \times 1 \text{ Å}^3$. The starting point of the random walk is shown in red, and the coordinates are given in Table I. The average diameters of the gramicidin channel and the three sections of the Na_v1.7 are shown in pink. d_{init} represents the distance between the initial point and the first xy-plane of the region to calculate diffusion along the z-axis. d_{total} represents the total distance from the topmost lattice site to the bottommost site, including the sites outside the region where diffusion is measured.

B. Continuous time random walk (CTRW) algorithm

We start by placing a single walker, representing Na⁺, on a lattice site. This initial site, shown in red in Fig. 2, is a couple of sites above what we set as the entrance of the pore. At each step, the walker waits on site i for a time τ_{wait}^i before moving to a neighboring site j . τ_{wait}^i is defined as⁶⁹

$$\tau_{\text{wait}}^i = -\frac{\ln(R)}{\sum_{j=1}^{nn} k_{ij}}, \quad (1)$$

where R is a random number between 0 and 1, k_{ij} is the rate at which the walker moves from site i to site j , and nn is the number of nearest neighbors. We only consider moves to a site j that is in the negative or positive x -, y -, or z -direction such that nn is 6 at maximum and less than 6 at the lattice boundaries (i.e., $nn \leq 6$). Equation (1) states that the lower the k_{ij} rates (slow dynamics), the longer the waiting time on site i . The total time the walker spends on the lattice is equal to the sum of the waiting times at the sites it visits.

After τ_{wait}^i , the walker moves to one of the neighboring sites j . The destination is picked according to⁶⁹

$$p_{ij} = \frac{k_{ij}}{\sum_j k_{ij}}, \quad (2)$$

where p_{ij} is the probability of moving to site j knowing that the walker is on site i .

1. Rate assignment for lattices with no disorder

In this case, we assign a single constant rate to each site, k^{ref} , that we multiply by a direction-specific factor, $b_{\pm x, y, z}$, to get k_{ij} .

In unbiased walks, this factor is set to unity for all directions such that $k_{ij} = b_{\pm x, y, z} k^{\text{ref}} = k^{\text{ref}}$ for all i and j .

In biased walks, the multiplicative factor is also set to unity for the positive and negative x - and y -directions, but varies in the z -direction such that

$$k_{i(\pm x, y)} = k^{\text{ref}}, \quad k_{i(-z)} = b_{-z} k^{\text{ref}}, \quad \text{and} \quad k_{i(+z)} = b_{+z} k^{\text{ref}}. \quad (3)$$

For the majority of the results presented in this paper, $b_{-z} = 2$ and $b_{+z} = 0.5$, but we also investigated scenarios where ($b_{-z} = 2$, $b_{+z} = 1$), ($b_{-z} = 3$, $b_{+z} = 1$), ($b_{-z} = 3$, $b_{+z} = 1/3$), and finally ($b_{-z} = 4$, $b_{+z} = 1$). Although the specific b_{-z}/b_{+z} ratios are arbitrary and chosen to illustrate how our algorithm performs under these conditions, the introduction of bias in general accounts for the combined effect of concentration gradient, transmembrane potential, small molecules, and other factors that drive the diffusion of ions through a channel in real systems.¹⁰

2. Rate assignment for lattices with static disorder

In this case, we draw a rate for each site, k_i , from a probability distribution $\mathcal{D}(k^{\text{ref}}, \sigma)$, with a mean k^{ref} and a standard deviation σ . In this paper, we investigate the influence of σ , which we take to be an increasing fraction of k^{ref} : $\sigma = m k^{\text{ref}}$, where $m = 0.01, 0.1, 0.25, 0.5, 0.75$, or 1.0 . In Fig. 3, we show the rate variations over a slice of the Na_v1.7 lattice when $\sigma = k^{\text{ref}}$.

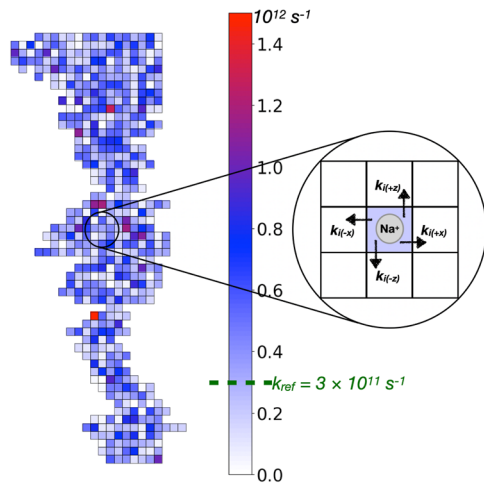


FIG. 3. Heat map of the rate variations over the sites in a slice of the Na_v1.7 lattice. The rates were drawn from $\mathcal{D}(k^{\text{ref}}, \sigma = k^{\text{ref}})$ using the Gaussian approach as described in the text. $k^{\text{ref}} = 3 \times 10^{11} \text{ s}^{-1}$.

The distribution $\mathcal{D}(k^{\text{ref}}, \sigma)$ is realized in two ways: (i) starting from a log-normal distribution (using `NUMPY.RANDOM.LOGNORMAL`⁷⁰), $\mathcal{L}_N(\mu_{\mathcal{L}_N}, \sigma_{\mathcal{L}_N})$ with

$$\mu_{\mathcal{L}_N} = \ln(k^{\text{ref}}) - \frac{1}{2} \ln\left(\frac{\sigma^2}{(k^{\text{ref}})^2} + 1\right), \quad (4a)$$

$$\sigma_{\mathcal{L}_N} = \sqrt{\ln\left(\frac{\sigma^2}{(k^{\text{ref}})^2} + 1\right)}, \quad (4b)$$

and using the underlying Gaussian distribution or (ii) starting from a Gaussian distribution (using `NUMPY.RANDOM.NORMAL`⁷⁰) $\mathcal{G}(\mu_{\mathcal{G}}, \sigma_{\mathcal{G}})$ with

$$\mu_{\mathcal{G}} = k^{\text{ref}} - \frac{1}{2} \frac{\sigma^4}{(k^{\text{ref}})^3}, \quad (5a)$$

$$\sigma_{\mathcal{G}} = \sigma, \quad (5b)$$

re-drawing from $\mathcal{G}$ until a positive value is obtained for k_i . Histograms of both the distribution $\mathcal{D}$ obtained from the log-normal ($\mathcal{D}^{\text{LN}}$) and the Gaussian ($\mathcal{D}^{\text{G}}$) distributions are shown in Figs. S3 and S4 along with a table of their means and standard deviations compared against the desired means and standard deviations (Table S1).

Bias is introduced similarly to lattices with no disorder, where

$$k_{i(\pm x, y)} = k_{\pm x, y}, \quad k_{i(-z)} = b_{-z} k_{-z}, \quad \text{and} \quad k_{i(+z)} = b_{+z} k_{-z}. \quad (6)$$

3. Rate assignment for lattices with dynamical disorder

In this case, we vary the rates over space (i.e., lattice sites) and time (i.e., random walk moves). As such, the walker faces a different set of rates if it returns to the same point in the lattice at a later time.

In unbiased walks, we draw these rates from $\mathcal{D}(k^{\text{ref}}, \sigma)$ for each site after each move.

In biased walks, we draw the rates for the positive and negative x - and y -directions from $\mathcal{D}(k^{\text{ref}}, \sigma)$ for each site after each move, while we draw the rates for the positive and negative z -directions from $\mathcal{D}(b_{\pm z} k^{\text{ref}}, \sigma)$.

In all these cases (e.g., lattices with no disorder, lattices with static disorder, and lattices with dynamic disorder), k^{ref} is estimated from the experimental apparent tissue diffusion coefficient of sodium, D^{exp} , as follows:^{71,72}

$$k^{\text{ref}} = \frac{6D^{\text{exp}}}{a^2}, \quad (7)$$

where $a = 1 \text{ \AA}$ and $D^{\text{exp}} = 5 \times 10^{-6} \text{ cm}^2/\text{s}$.⁷³ It yields $k^{\text{ref}} = 3 \times 10^{11} \text{ s}^{-1}$. The dependence of the diffusion coefficient obtained from our simulations on the site size is shown in Fig. S1 for 2D lattices.

C. Calculation of the diffusion coefficient

The coordinates of every site visited by the walker ($\vec{r}_i$) are stored, along with the corresponding waiting times (τ_{wait}^i). The velocity of the walker after each step in trajectory l is then calculated as the total distance traveled divided by the total time elapsed, according to

$$v_l^i = \frac{|\vec{r}_l^i - \vec{r}^0|}{\sum_{s=0}^i \tau_{\text{wait}}^{ls}}, \quad (8)$$

where v_l^i is the velocity of the walker on site i in trajectory l , $\vec{r}_l^i$ is the coordinates of site i in trajectory l , $\vec{r}^0$ is the coordinates of the initial site i (same for all trajectories), and τ_{wait}^{ls} is the waiting time at site i in trajectory l . The Velocity AutoCorrelation Function (VACF) for each trajectory l is then calculated using $\vec{v}^l(t) \cdot \vec{v}^l(t + t_0)$, where $\vec{v}^l(t)$ is the vector velocity for trajectory l whose components are v_l^i and t_0 is the time lag. t_0 is set to be the time taken to complete 10% of the total number of steps in that trajectory. The total number of steps is the number of steps required for the walker in that trajectory to exit the pore, or 500 if the walker did not exit the pore. Note that we chose our CTRW simulations to be 500 steps because the walker almost always exits the pore at that time. The VACF is then averaged over the number of trajectories to get the mean VACF $\langle v(t) \cdot v(t + t_0) \rangle_l$. The elements of the resulting array (containing the averaged VACF at each step) are then summed to get D_{VACF} , according to⁷¹

$$D_{\text{VACF}} = \int_0^\infty \langle \vec{v}(0) \cdot \vec{v}(t) \rangle dt. \quad (9)$$

The domain of $\vec{r}_i$ (and, by extension, v_i) can be modified to contain only certain sections of the pore or the entire pore (at maximum d_{diff} , as shown in Fig. 2) to get the corresponding D_{VACF} . In this work, we average over 5000 trajectories of 500 steps each.

For comparison, we also compute the Walker diffusion coefficient from its Mean Square Displacement (MSD) as

$$D_{\text{MSD}} = \lim_{t \rightarrow \infty} \frac{1}{2d} \frac{\langle |\vec{r}_l^i - \vec{r}^0|^2 \rangle_l}{t}, \quad (10)$$

TABLE II. Diffusion coefficient and diffusion time (time taken to advance the length of diffusion d_{diff}) from CTRW run on the three lattices studied in this paper. Each number results from an average of over 5000 trajectories of 500 steps each, with no disorder.

System	b_{-z}	b_{+z}	D_{VACF} in $\text{cm}^2 \text{s}^{-1}$	t_{diff} in s
Open lattice	1	1	5.0×10^{-6}	27.5×10^{-11}
Gramicidin	1	1	2.2×10^{-6}	10.7×10^{-11}
$\text{Na}_v 1.7$	1	1	1.8×10^{-6}	33.4×10^{-11}
Open lattice	2	1/2	68.5×10^{-6}	9.7×10^{-11}
Gramicidin	2	1/2	39.0×10^{-6}	4.8×10^{-11}
$\text{Na}_v 1.7$	2	1/2	11.9×10^{-6}	25.1×10^{-11}

where $|\vec{r}_t^j - \vec{r}^0|$ is the displacement of the walker from its original position and t is the time of the simulation (i.e., the sum of the waiting times up to the walker's current position).

III. RESULTS AND DISCUSSION

A. Validation of the methods

In this work, we use rate distributions of mean k^{ref} and varying standard deviation σ . We set $k^{\text{ref}} = 3 \times 10^{11} \text{ s}^{-1}$, using the continuum limit of the ideal random walk formalism with an experimental value of the Na^+ diffusion coefficient in cells [D^{exp} in Eq. (7)]. Here, an ideal random walk means that all directions are equiprobable and the time and length of the steps are constant. By design, we should recover this value of D^{exp} when running our CTRW algorithm on the open lattice with no disorder and no applied bias. In Table II, we give the value of the diffusion coefficients computed according to Eq. (9), using 5000 CTRW simulations for each of the open, gramicidin, and $\text{Na}_v 1.7$ lattices.

We find that $D_{\text{VACF}} = D^{\text{exp}} = 5 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$ in the open lattice, validating our approach. Furthermore, we see that D_{VACF} decreases with increasing complexity of the lattice and, more importantly, irrespective of the lattice volume or the length of diffusion. Indeed, the gramicidin lattice is the smallest overall, with a volume of $553 \text{ \AA}^3$ and a length of diffusion of $d_{\text{diff}} = 14 \text{ \AA}$. Both the open and $\text{Na}_v 1.7$ lattices have $d_{\text{diff}} = 44 \text{ \AA}$, but the volume differs greatly with $300\,763$ and $5499 \text{ \AA}^3$, respectively. In unbiased walks, t_{diff} scales with d_{diff} , but D_{VACF} does not scale as the inverse of t_{diff} . For example, D_{VACF} in the gramicidin lattice is half of that in the open lattice, despite t_{diff} in the gramicidin lattice being a third of that in the open lattice. This suggests that it is the number of available nearest neighbors that dictates the magnitude of D_{VACF} , with the more complex lattices (i.e., lattices with the most sites where $nn < 6$) exhibiting the lowest diffusion kinetics. This makes sense given Eq. (1), where a lower number of nearest neighbors will result in a longer waiting time when all rates are equal. When applying bias in the z -direction, t_{diff} no longer scales linearly with d_{diff} , but D_{VACF} still decreases with increasing lattice complexity.

Note that another consequence of the complexity of the lattices is that the mean-square displacement of the walker does not scale linearly with time, especially when applying disorder (see Figs. S5 and S6 in supplementary material). This means that Eq. (10) is not valid in those cases, unlike Eq. (9), which we use throughout this paper.

B. Influence of static and dynamic disorder

In Fig. 4, we present the diffusion coefficients, D_{VACF} , computed from CTRW simulations where disorder has been introduced via site-dependent transition rates. In this case, each site is assigned a different rate, drawn from a distribution $\mathcal{D}(k^{\text{ref}}, \sigma)$ of mean k^{ref} and standard deviation σ . As explained in Sec. II, we realized $\mathcal{D}$ in two ways: (i) via a log-normal distribution, in which case the final rate distribution is labeled $\mathcal{D}^{\text{LN}}$ [Figs. 4(a) and 4(b)], and (ii) via a normal distribution, in which case the final rate distribution is labeled $\mathcal{D}^{\text{G}}$ [Figs. 4(c) and 4(d)]. Using both of these distributions, we vary the magnitude of the disorder by increasing σ from $0.01k^{\text{ref}}$ (no disorder limit) to k^{ref} (maximal disorder). Finally, we compare D_{VACF} when the disorder is static (i.e., the rates at each site are drawn from the distribution at the beginning of the simulation only) and dynamic (i.e., the rates at each site are drawn from the distribution at every time step). The raw data associated with Fig. 4 and the 95% confidence interval over five sets of 5000 CTRW simulations are given in Tables S2–S5.

We observe that the type (i.e., static or dynamic) of disorder does not change D_{VACF} significantly in unbiased walks [Figs. 4(a) and 4(c)]. However, the magnitude of the disorder, for example, using the data from CTRW runs with $\mathcal{D}^{\text{LN}}$, will result in a drop in D_{VACF} of 25%, 32%, and 33% for the open, gramicidin, and $\text{Na}_v 1.7$ lattices, respectively. Similar observations can be made using the data from CTRW runs with $\mathcal{D}^{\text{G}}$. When a bias is applied in the z -direction, both the type and the magnitude of the disorder influence D_{VACF} . Note that the CTRW simulations did not converge at large dynamic disorder in Fig. 4(d), as it is more difficult to draw a positive rate from $\mathcal{G}(k^{\text{ref}}, \sigma = k^{\text{ref}})$. Nevertheless, we see that D_{VACF} is significantly reduced as the disorder increases in all lattices, but only when the disorder is static. When the disorder is dynamic, D_{VACF} remains approximately constant for all three lattices. This means that at equal disorder, the diffusion kinetics are faster with dynamic disorder than with static disorder. Overall, these results indicate that when there is no bias, the walker can avoid areas with lower rates by visiting neighboring sites in other directions. However, when bias is applied, the walker is forced through sites with lower rates along the z -direction, which reduces D_{VACF} . This is recovered with dynamic disorder, where the renewal of the rates on each site allows the walker to effectively escape areas that were previously assigned low rates. This is significant because it implies that using the potential of mean force in traditional phenomenological models overlooks the effect protein motions occurring on the scale of ion diffusion (modeled via our dynamic disorder) have on diffusion kinetics.

C. Influence of the magnitude of the bias

Next, we investigate the influence of the magnitude of the bias on the absolute normalized difference in D_{VACF} , defined as

$$\Delta D_{\text{VACF}}^{\text{norm}} = \frac{|D_{\text{VACF}}(\text{dynamic}) - D_{\text{VACF}}(\text{static})|}{\frac{1}{2}(D_{\text{VACF}}(\text{dynamic}) + D_{\text{VACF}}(\text{static}))}, \quad (11)$$

where $\Delta D_{\text{VACF}}^{\text{norm}}$ is the absolute normalized difference between the diffusion coefficient computed at maximal dynamic disorder, $D_{\text{VACF}}(\text{dynamic})$, and the diffusion coefficient computed at maximal static disorder, $D_{\text{VACF}}(\text{static})$. In our simulations, the bias

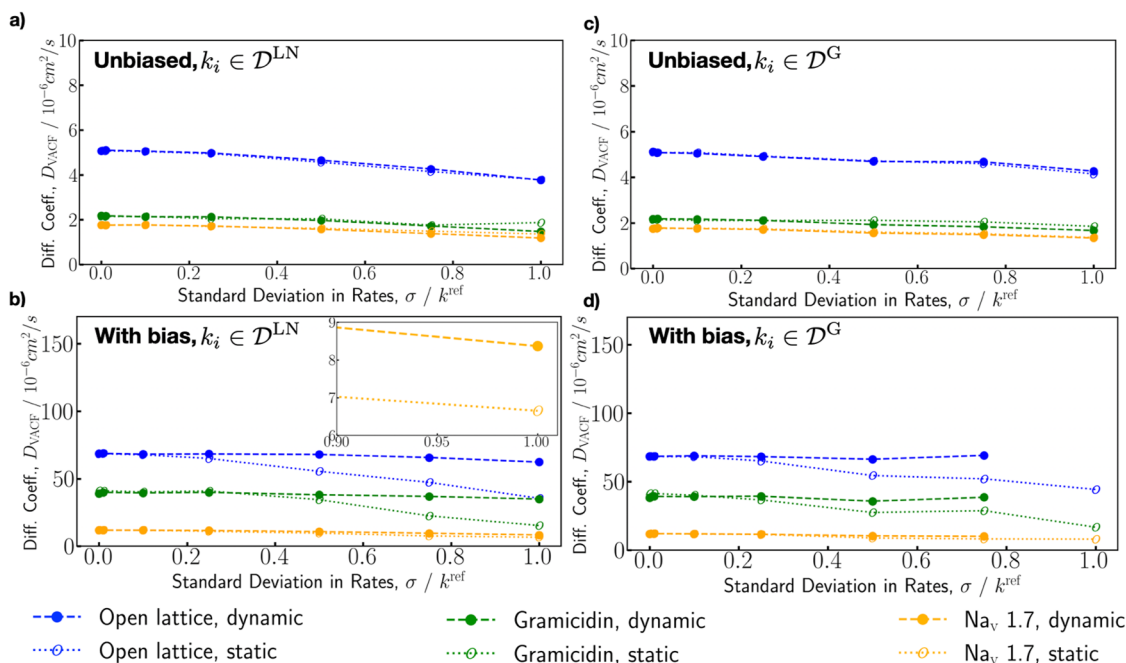


FIG. 4. Diffusion coefficient as a function of σ . D_{VACF} is computed according to Eq. (9) from an average of 5000 CTRW simulations of 500 steps each, repeated five times (25 000 simulations in total). (a) The rates are drawn from $\mathcal{D}^{LN}$, and no bias is applied. (b) The rates are drawn from $\mathcal{D}^{LN}$, and a bias $[(b_{-z}, b_{+z}) = (2, 1/2)]$ is applied. (c) The rates are drawn from $\mathcal{D}^G$, and no bias is applied. (d) The rates are drawn from $\mathcal{D}^G$, and a bias $[(b_{-z}, b_{+z}) = (2, 1/2)]$ is applied.

accounts for the combined effect of concentration gradient, transmembrane potential, small molecules, and other factors that drive the diffusion of ions through a channel in real systems.¹⁰ In Fig. 5, we show ΔD_{VACF}^{norm} plotted for different pairs of b_{-z} and b_{+z} .

We observe that applying a bias in the z -direction always results in an increased ΔD_{VACF}^{norm} compared to the unbiased case. However, ΔD_{VACF}^{norm} seems to reach a plateau relatively quickly. For the open and gramicidin lattices, this plateau is $\Delta D_{VACF}^{norm} = 75\%$, while it is 50% for

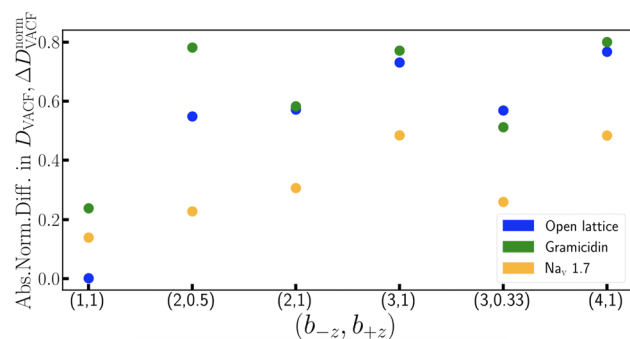


FIG. 5. Absolute normalized difference between the diffusion coefficient computed at maximal dynamic and static disorder, ΔD_{VACF}^{norm} , plotted for different pairs of b_{-z} and b_{+z} . Each data point is the average of 5 sets of 5000 CTRW simulations of 500 steps each using $\mathcal{D}^{LN}$.

the Na_v1.7 lattice. This sets a limit on how much diffusion kinetics can be accelerated through molecular motion and dynamics in a disordered, constricted landscape.

D. Local variations in diffusion kinetics in the Na_v1.7 lattice

Finally, we look at local variations in D_{VACF} in the more complex Na_v1.7 lattice. To do that, we separate the pore into a selectivity filter, central cavity, and inactivation gate regions, as described in Sec. II and shown in Fig. 6. The raw data and the 95% confidence interval over the five sets of 5000 CTRW simulations are given in supplementary material (Tables S6–S9). We see that in unbiased walks, the diffusion kinetics is the slowest in the selectivity filter, followed closely by the central cavity. Note that the type of disorder does not influence this kinetics, as was observed in Fig. 4 for the entire pore. Interestingly, D_{VACF} is the highest in the inactivation gate region.

In biased walks at low disorder, the diffusion kinetics in the selectivity filter are still the slowest, but the kinetics in the central cavity are now faster than in the inactivation gate region. Note that this is true regardless of the distribution used for the rates. As the disorder increases, at around $\sigma = 0.2k^{ref}$ for $\mathcal{D}^{LN}$ and $\sigma = 0.4k^{ref}$ for $\mathcal{D}^G$, we again see that D_{VACF} is higher with dynamic disorder than static disorder for all regions. This means that our previous observations hold at a smaller length scale, suggesting that the motion of individual residues is likely enough to significantly affect the permeation kinetics of Na⁺ in pores with complex geometries.

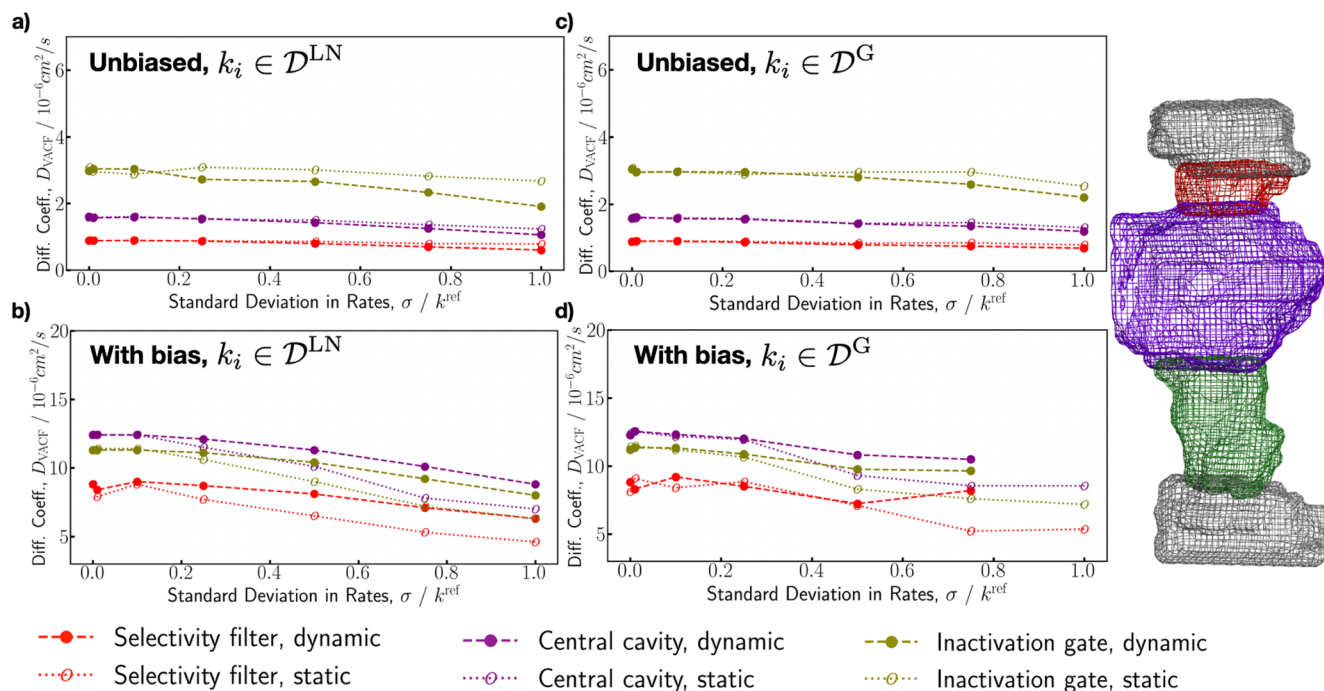


FIG. 6. D_{VACF} for each region computed according to Eq. (9) from an average of 5000 CTRW simulations of 500 steps each, repeated five times (25 000 simulations in total). (a) The rates are drawn from $\mathcal{D}^{\text{LN}}$, and no bias is applied. (b) The rates are drawn from $\mathcal{D}^{\text{LN}}$, and a bias $[(b_{-z}, b_{+z}) = (2, 1/2)]$ is applied. (c) The rates are drawn from $\mathcal{D}^{\text{G}}$, and no bias is applied. (d) The rates are drawn from $\mathcal{D}^{\text{G}}$, and a bias $[(b_{-z}, b_{+z}) = (2, 1/2)]$ is applied.

IV. CONCLUSION

In this paper, we present a CTRW model of ion diffusion through channels. Our walks run on 3D lattices that map out the specific geometry of a given channel pore—gramicidin and $\text{Na}_v1.7$ in this case. Our CTRW algorithm revolves around the introduction of a walker waiting time at each site that we define as a function of the transition rates to neighboring sites. We constructed our model such that slow dynamics (i.e., low transition rates to neighboring sites) yield long waiting times and vice versa. Furthermore, we explicitly incorporated disorder by drawing these transition rates at each site from distributions with varying standard deviations (static disorder). These rate distributions mimic the heterogeneous and dynamic potential energy landscape through which the ion diffuses in real systems. We found that diffusion kinetics are slowed down by this heterogeneity when a bias is applied along the principal direction of the channel pore and that the ion is forced to diffuse through areas of lower transition rates. Consequently, we also find that updating these rates periodically (dynamic disorder) helps to recover diffusion kinetics as it enables the walker to avoid areas with low transition rates. Our findings suggest that proteins' structural and dynamical heterogeneity significantly modulate the kinetics of ion diffusion through channels. Consequently, phenomenological models based on a mean field approximation of the force exerted on the ion miss important physics and cannot fully rationalize the observed behavior of channels at the microscopic scale. In its current form, our model accounts for the specific geometry of the system but not the chemistry. Our rate distributions were designed to reproduce the average

experimental diffusion rate, but they do not account for the chemical nature of the channel at the molecular scale. The values for the bias were also chosen for numerical experiments as a demonstration of the principle. In the future, we plan on addressing these limitations by parameterizing our CTRW algorithm using data from MD simulations to integrate the information obtained at the atomistic scale.

SUPPLEMENTARY MATERIAL

A document containing supplementary material accompanies this text. It contains the following data: The plot of the diffusion coefficient as a function of site size. The number of pore sites in the (xy) plane as a function of position along the (z) axis for the three lattices studied here. Histograms of the distributions $\mathcal{D}^{\text{LN}}$ and $\mathcal{D}^{\text{G}}$ and a table with their means and standard deviations. Plots of the diffusion coefficient as a function of time, computed from the mean-square-displacement and the velocity autocorrelation function. Tables containing the data used to make the graphs are in Figs. 4–6.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Ronnie Mondal: Data curation (lead); Formal analysis (supporting); Investigation (supporting); Methodology (equal); Visualization (lead); Writing – original draft (supporting). **Valerie Vaissier Welborn:** Conceptualization (lead); Formal analysis (lead); Funding acquisition (lead); Investigation (lead); Methodology (equal); Validation (lead); Writing – original draft (lead).

DATA AVAILABILITY

The data and codes that support the findings of this study are openly available in the public GitHub repository at https://github.com/WelbornGroup/CTRW_Project.

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